N-gram Activity Sheet

Backoff • Deleted Interpolation • Entropy and Perplexity

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Corpus used: "the cat sat on the mat . the dog sat on the rug . the cat ate fish ."

Calculation note: The corpus has 16 word tokens (excluding the three sentence-ending full stops). For unigram probabilities, the denominator is therefore 16. For Activity 1, the backoff score is taken to be the relative-frequency probability at the first available level, multiplied by 0.4 for each level of fallback. This makes the assumed scoring convention explicit.

Activity 1: Backoff (3 marks)

Backoff uses the highest-order n-gram available. If a trigram is not observed, back off to a bigram; if that is also unavailable, back off to a unigram. Apply the 0.4 penalty for each fallback.

a) "the cat sat"

The trigram "the cat sat" occurs once, and the bigram "the cat" occurs twice.

P(sat | the cat) = C(the cat sat) / C(the cat) = 1 / 2 = 0.5.

Answer: Score = 0.5; trigram level (no backoff needed).

b) "dog ate fish"

The trigram "dog ate fish" is not observed, and the bigram "dog ate" is not observed. Back off to the unigram "fish", which occurs once among 16 word tokens.

P(fish) = 1 / 16 = 0.0625.

Score = $0.4^2 \times (1 / 16) = 0.16 / 16 = 0.01$.

Answer: Score = 0.01; unigram level (two fallback steps).

c) "dog sat fish"

The trigram "dog sat fish" is not observed, but the bigram "dog sat" occurs once. Using the bigram conditional probability, P(sat | dog) = C(dog sat) / C(dog) = 1 / 1 = 1.

Apply one fallback penalty: score = 0.4 × 1 = 0.4.

Answer: Score = 0.4; bigram level (one fallback step).

Activity 2: Deleted Interpolation (6 marks)
Use the interpolation formula $P(w \mid u,v) = \lambda_1 P(w) + \lambda_2 P(w \mid v) + \lambda_3 P(w \mid u,v)$, where λ_1, λ_2 and λ_3 are the unigram, bigram and trigram weights respectively.

Part A (2 marks)
Given $\lambda_1 = 0.1$, $\lambda_2 = 0.3$, $\lambda_3 = 0.6$, find P(sat | the cat).

From the corpus: P(sat) = 2/16; P(sat | cat) = 1/2; P(sat | the cat) = 1/2.

P(sat | the cat) = 0.1(2/16) + 0.3(1/2) + 0.6(1/2)

= 0.0125 + 0.15 + 0.30 = 0.4625.

Answer: P(sat | the cat) = 0.4625.

Part B (4 marks)
Held-out sentence: "the dog ate fish". Compare Set 1 λ = (0.1, 0.3, 0.6) and Set 2 λ = (0.3, 0.6, 0.1).

Trigram 1: "the dog ate"

P(ate) = 1/16; P(ate | dog) = 0; P(ate | the dog) = 0.

Set 1: 0.1(1/16) + 0.3(0) + 0.6(0) = 0.00625.

Set 2: 0.3(1/16) + 0.6(0) + 0.1(0) = 0.01875.

Trigram 2: "dog ate fish"

P(fish) = 1/16; P(fish | ate) = 1; P(fish | dog ate) = 0.

Set 1: 0.1(1/16) + 0.3(1) + 0.6(0) = 0.30625.

Set 2: 0.3(1/16) + 0.6(1) + 0.1(0) = 0.61875.

Sum of natural logarithms:

Set 1: $\ln(0.00625) + \ln(0.30625) \approx -5.0752 - 1.1835 = -6.2587$.

Set 2: $\ln(0.01875) + \ln(0.61875) \approx -3.9766 - 0.4801 = -4.4567$.

Conclusion: Choose Set 2 because its total log-likelihood (approximately −4.4567) is higher (less negative) than Set 1's (approximately −6.2587). Therefore, Set 2 assigns a higher probability to the held-out sentence under these interpolation calculations.

Activity 3: Entropy and Perplexity (6 marks)
Entropy formula: $H = -\Sigma\, p \log_2(p)$. Perplexity is $PP = 2^H$.

Part A: Weather (2 marks)
Sunny: p = 0.5; Cloudy: p = 0.25; Rainy: p = 0.25.

$H = -[0.5 \log_2(0.5) + 0.25 \log_2(0.25) + 0.25 \log_2(0.25)]$

$= -[0.5(-1) + 0.25(-2) + 0.25(-2)] = 1.5$ bits.

$PP = 2^{1.5} \approx 2.828$.

Answer: Entropy = 1.5 bits; perplexity ≈ 2.828.

Part B: Next-word prediction (4 marks)

Words immediately following "the" in the corpus are: cat, mat, dog, rug, cat. Thus, "cat" occurs twice and "mat", "dog" and "rug" each occur once, for five total occurrences.

Next word	Count	Probability
cat	2	2/5 = 0.4
dog	1	1/5 = 0.2
mat	1	1/5 = 0.2
rug	1	1/5 = 0.2

$H(\text{after "the"}) = -[0.4 \log_2(0.4) + 3 \times 0.2 \log_2(0.2)]$

$\approx -[0.4(-1.3219) + 0.6(-2.3219)] \approx 1.9219$ bits.

$PP(\text{after "the"}) = 2^{1.9219} \approx 3.79$.

After "sat", the next word is "on" in both occurrences. Therefore $P(\text{on} \mid \text{sat}) = 1$, so $H(\text{after "sat"}) = -1 \log_2(1) = 0$ bits.

Answer: After "the": entropy ≈ 1.9219 bits and perplexity ≈ 3.79. After "sat": entropy = 0 bits.

Summary of Final Answers

Question	Final answer
1(a)	0.5 — trigram
1(b)	0.01 — unigram, two fallback steps
1(c)	0.4 — bigram, one fallback step
2(A)	P(sat | the cat) = 0.4625
2(B)	Set 1 log sum ≈ −6.2587; Set 2 ≈ −4.4567; choose Set 2
3(A)	Entropy = 1.5 bits; perplexity ≈ 2.828
3(B), after "the"	Entropy ≈ 1.9219 bits; perplexity ≈ 3.79
3(B), after "sat"	Entropy = 0 bits